Egocentric Human Demonstrations for a Bimanual Robot

From a Hand-Worn Gripper to Robot Policies

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Public repository: `github.com/physical-ai-robot-learning/egocentric-cube-stacking`.

***Abstract*—We study how far egocentric human demonstrations, recorded with HandUMI, an open-source hand-worn UMI-style gripper designed and published by BrikHMP18 (not our design), can be turned into training data for policies on a reBot B601 dual-arm robot. One report covers the whole chain: the capture rig, calibration, monocular wrist poses with three metric-scale sources, two generations of conversion into robot training data, a 0.9B-parameter X-VLA model, offline and diagnostic evaluation, and deployment. *Stacking.* We collected 349 bimanual demonstrations of three-cube stacking in six balanced orders. Retargeted to robot joints, they yield 561 bimanual segments; a model fine-tuned on them predicts the direction of held-out human motion on moving samples (cosine 0.90 at 1.2 s, joint error 13–32% below a zero-motion predictor) but drifts during static phases, and swapping the order instruction does not make predictions more accurate. Converting whole episodes instead to the robot's own relative Cartesian action contract keeps 330 episodes with no feasibility filter. With that contract the remaining gap is visual: an ego-only model gives no useful action direction on robot frames (median cosine -0.05/-0.03 versus $+0.81$/$+0.77$), and the gap follows the wrist images. As an initialization for robot fine-tuning, ego pretraining lowers the early training loss (3.9× at the start on 312 episodes, about half on 30 episodes), and the advantage disappears as the robot data is fitted. An ego-initialized model fine-tuned on 312 robot episodes completed full three-cube stacks in closed loop; trials were not counted. *Single-arm approach.* For slow reaching motions IMU scale is unobservable; a cube of known size and a table plane replace it, and a taped origin touched by both robot and human gives a shared task frame (collinearity 0.38 mm). Models trained on 80–150 takes overfit from the first checkpoint; on the robot they move towards the cube from nearby starts. We report no success rate anywhere, and we state which claims this data can and cannot support.**

I. Introduction

Collecting robot demonstrations requires a robot, an operator and task-specific teleoperation infrastructure. Egocentric human demonstrations can be collected faster, with less hardware and in natural settings, and hand-worn grippers such as UMI [1] make the human's end-effector look like the robot's. Whether such data transfers to a specific robot depends on a chain of steps that is rarely audited end to end: sensor calibration, metric pose recovery from a moving wrist camera, conversion into the robot's action space, and the filtering that decides which human motion is kept. Each step can silently change what a policy learns.

This report documents that chain on real data for one robot and two tasks, and asks two narrow questions. *(Q1) After all processing, how much robot-executable action structure does a pretrained vision-language-action (VLA) model learn from egocentric human demonstrations of bimanual stacking, and does it use the task-order instruction? (Q2) What has to be true of the data for the simplest transferable behaviour, a single-arm approach to a cube, to move the robot towards the cube?*

Our contributions are: a build record of the capture hardware with our changes to the upstream design (Sec. III); a 349-episode bimanual stacking dataset and two single-arm approach datasets with full calibration (Secs. III–IV); pose recovery with three independent scale sources and the regimes in which each fails (Sec. V); two conversions into robot training data, joint-space retargeting and a Cartesian contract shared with the robot data, with what each gate removes (Sec. VI); an offline evaluation of joint-space stacking with a motion-gated analysis and a prompt-swap test (Sec. IX); transfer diagnostics that localize the human-to-robot gap, and initialization comparisons at 312 and 30 robot episodes (Sec. X); a calibration chain that puts single-arm human data into the robot's task frame (Sec. XI); and the deployment settings and corrections that turned out to matter (Secs. XII–XIII).

We are explicit about scope. Offline evaluations are on held-out *episodes* that share operator, scene and all stacking orders with training; cube layouts were not recorded. Robot runs were exploratory and outcomes were not logged, so there is no success rate. Sec. XV lists the experiments needed for the claims we do not make.

II. Related Work

Portable demonstration interfaces. UMI [1] records hand-held gripper demonstrations with a wrist fisheye camera and recovers poses with ORB-SLAM3 against a per-scene map; DexCap [2] targets dexterous hands, and YUBI [3] a finger-driven gripper. Our device is HandUMI [4], an open-source hand-worn UMI variant for parallel-jaw grippers with a servo-encoder jaw; we replace its VR tracking with a wrist IMU. **Learning from egocentric humans.** EgoMimic [6] and MimicPlay [7] co-train robot policies on human egocentric video and robot data; large corpora such as Ego4D [8] motivate the scaling argument. These works report closed-loop gains; we audit what survives processing and what a model learns, which is a prerequisite for interpreting such gains. **Policies.**

Action-chunking imitation (ACT [9], Diffusion Policy [10]) and VLA models (OpenVLA [11], π_0 [12], X-VLA [13]) predict action chunks from images and language. We use X-VLA through LeRobot [14] because its flow-matching [15] head and soft prompts accommodate a new embodiment with a small dataset. **Pose recovery.** MASt3R-SLAM [16] provides dense monocular SLAM from a 3D reconstruction prior; we combine it with IMU, object and plane scale, and calibrate with Kalibr [17].

III. Hardware and Data Collection

Attribution. The capture device is *not our design*. HandUMI was designed and published as open-source hardware by BrikHMP18 (https://github.com/murobotics-ai/handumi-hw, Apache-2.0) [4], together with its software (https://github.com/murobotics-ai/handumi-sw) [5]. The repositories were first released on 20 August 2026 under `robonet-ai`, which now redirects. We built the hardware from upstream commit `e58de33`. Our own hardware work is limited to three modified parts, a wrist IMU added in place of the upstream VR tracking, and a fixed head camera. Our `umihand` repository is a build copy of the upstream files plus these additions, not an original design.

Upstream HandUMI. HandUMI is a hand-worn variant of UMI for robot arms with parallel-jaw grippers. It mounts on the thumb and the index and middle fingers and closes with a natural pinch; its gripper tips are detachable (AgileX Piper, ARX X5, Dream Gripper, WidowX AI and the UMI gripper), so one body serves several robots. Upstream records wrist SE(3) from a VR headset (PICO 4 Ultra or Meta Quest 3) and its controllers, held on the device by a printed controller support; jaw width directly from a Feetech servo encoder rather than from fiducials or image segmentation; and wrist-view video. The finger cradles were modelled on a 3D scan of the author's hand. The companion `handumi-sw` records, visualizes and retargets demonstrations. Upstream has since moved to an IMX335 camera and a per-pair BOM (commit `fc7462a`); we did not adopt that change. We forked `handumi-sw` at commit `33cc437` for an early ego-bimanual path (Quest 3S tracking, AprilTag end-effector tracking, Feetech jaw telemetry, a reBot B601 robot profile and tracking-quality gates); none of the demonstrations in this report were recorded with that path.

Build. The unit is a hand-support base with finger cradles, a fisheye camera support and a crank linkage (MR63ZZ bearings, LM4UU linear bearings on a 4 mm shaft) that couples the finger closure to the gripper tip and to a servo read as the jaw encoder. One unit is 12 printed parts; the package has 9 plates and 26 parts for both hands plus a generic tip, about 9 h and 250 g of PLA (0.20 mm layers, 2 walls, 20% infill, Bambu X1C), about 4 h and 110 g per hand. A fit-check plate (cradles and base, about 1 h) is printed first for each hand; tight holes are fixed with $+0.05$ to 0.1 mm XY hole compensation, not by scaling. The tip in the package is the AgileX-Piper tip, a temporary tip for mechanism and pinch tests; no reBot-specific tip has been designed. The upstream BOM is about US$110 per unit.

Sensors and our changes. Each unit carries an Arducam 160° fisheye USB camera (1920×1080 MJPG, 30 fps) and a Feetech STS3215 servo read as the jaw encoder (4096 ticks/rev, polled at 100 Hz) through a Waveshare serial-bus adapter (CH343, 1 Mbaud, servo id 1, which needs its 12 V supply before the servo answers). We changed three things. (i) VR tracking is replaced by the wrist camera plus an IMU, so the controller support is replaced by our connector. (ii) Each unit gets an ICM-42688-P IMU on a Teensy 4.1 over SPI. The sensor has no 400 Hz rate, so it runs at 200 Hz (measured 201.6 and 197.9 Hz, each board having its own oscillator); the data-ready interrupt time-stamps each sample on the device, packets (40-byte payload, CRC-16) go over USB CDC with a 1 Hz status packet reporting dropped samples, and the host fits the device-to-host clock offline so USB jitter does not enter the timeline. (iii) A fixed Logitech C922 head camera at 640×480 and 30 fps records the table and is the global view given to the policy; its autofocus is off because it changed f_x between sessions (417, 531, 501 px). Three parts were edited in Shapr3D (Fusion 360 and FreeCAD were tried and rejected; Fig. 2): a new left controller-support connector (8 September, 85.7×21.4×42.1 mm, printed twice) whose clamp plate was then lengthened by 10 mm (14 September, depth 21.4→31.4 mm; 15 mm and three z+30 mm variants were rejected), a camera mount with a second window, and a servo cover with a wider cable opening.

Device identity. Both Arducams report the same USB serial (`UC684`), so the recorder matches them by product name and refuses to record unless exactly one device matches each name; IMUs are matched by Teensy USB serial and jaws by adapter serial plus servo id, never by port order. The Teensy must be flashed as USB Serial; a RawHID build shows no port and looks dead. One jaw crosses the encoder's 4095→0 seam, so jaw normalization is circular. A 12.2 V supply sets the servo's "voltage" error bit against its 12.0 V limit; this is telemetry, not a fault. An RGB-D head camera was dropped because it disrupted the wrist streams.

Recording. An episode directory holds one MP4 per camera, an MCAP file with IMU samples (device and host timestamps), gripper samples (wrapped and unwrapped ticks), per-frame metadata, events and clock anchors, and metadata files; it is renamed from `.incomplete` to `.complete` only when closed. All streams are stamped with the host monotonic clock (IMU at receive time, gripper at the midpoint of the serial transaction, frames at capture).

Collection protocols. An AUTO loop (RESET → ARMING → HOLD → EPISODE → STOP) drives each take with spoken cues. ARMING waits for stillness (in 0.25 s windows, mean gyro magnitude below 6°/s, accelerometer-magnitude standard deviation below 0.35 m/s^2, largest sample gap below 25 ms, for a continuous 1 s); a hold after recording starts gives the visual–inertial initialization window. A take that passes preliminary QA (no device or recorder error, no empty stream, at least 25 fps, gripper sampled at 20 Hz or more, not frozen,

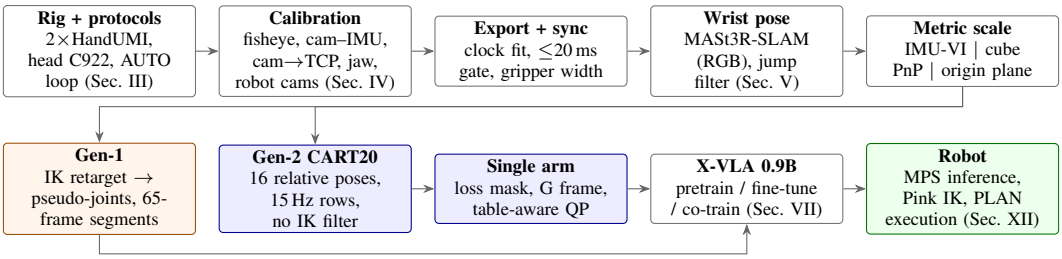


Fig. 1. Pipeline. Orange: generation 1 (joint space); blue: generation 2 (Cartesian) and its single-arm variant (Sec. VI); green: deployment. Every stage writes a manifest and is followed by a gate that accepts a unit (episode, segment, row or take) or rejects it with a logged reason.

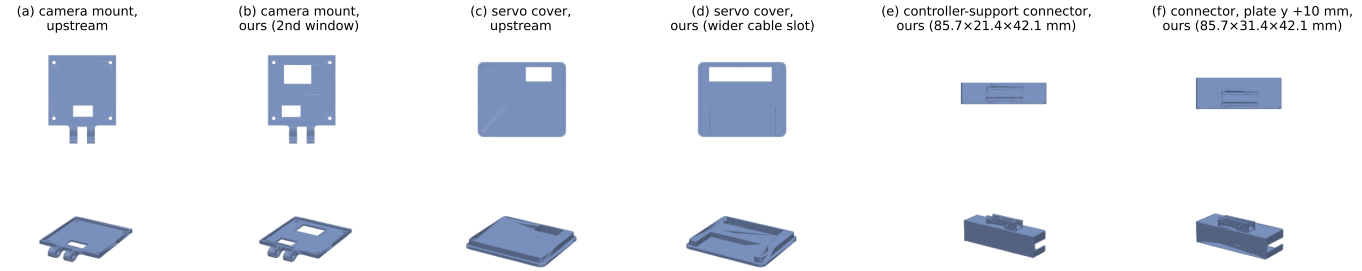


Fig. 2. Our changes to the upstream HandUMI parts (renders of the printed meshes). (a,b) camera mount: a second window cut above the existing one; (c,d) servo controller cover: the cable opening widened; (e) the new left controller-support connector that replaces the upstream VR-controller support; (f) its clamp plate lengthened by 10 mm, the current version.

within its calibrated span and travelling at least 5% of its range) is kept automatically.

- *Stacking*: three 50 mm cubes (red, blue, purple) are stacked on a fixed plate in one of six orders named by the instruction, e.g. "Stack the blue cube on the bottom, purple cube in the middle, and red cube on the top, on the plate." Takes last 20 s with a 10 s reset in which the operator re-places the cubes by hand. The next order is the least-collected one (target 50 per order), so orders are *balanced*, not randomized. The layout varies through the manual reset around a nominal layout matched to the robot workspace but was not recorded; there is no per-episode layout label.
- *Robot-like* (HRL80): 30 s stacking takes with live feedback; wrist angular speed and acceleration on a 15 Hz grid are compared with the 95th percentiles of robot teleoperation data (0.996 rad/s, 4.413 rad/s^2), and a take is reviewed if more than 10% of ticks exceed them or fewer than two grasps occur.
- *Approach* (HRA_red): right hand only, "approach the red cube", 10 s takes.
- *Approach with origin* (HRA_A100): as above, but each take starts with the jaw tip held on a taped origin shared with the robot (Sec. XI).

Datasets. Table I lists every human and robot dataset used in this report. One operator recorded all human data. The 349 stacking episodes come from 15 sessions over three days (15–17 September 2026), 2.20 h in total (median episode 23.05 s), per order RBP 64, RPB 62, BRP 61, BPR 56, PRB 55, PBR 51. Episodes from ten early sessions (55 episodes) are tracked as a separate provenance domain ("early") from the main 294 because their gripper anchors were derived afterwards; after filtering the two domains behave alike. All robot data are teleoperated demonstrations of the same six-order stacking task on the reBot B601.

TABLE I
DATASETS. HUMAN DATA WERE RECORDED WITH HANDUMI; ROBOT DATA ARE TELEOPERATED REBOT B601 EPISODES OF THE SAME STACKING TASK.

Name	Content
Stacking	349 bimanual episodes, 6 balanced orders, 15 sessions, 15–17 Sep.
HRL80	60 robot-like stacking episodes (10 per order), one session
HRA_red	200 right-hand approach takes, 3 Oct.
HRA_A100	107 approach takes with a taped origin, 6 Oct.; 27 more on 7 Oct.
R150 / R90	150 / 90 robot episodes (Gen-1 fine-tuning)
R312c	312 robot episodes, 174,012 frames (Gen-2 fine-tuning)
R30	episodes 150–179 of R312c, 17,600 frames, 5 per order
R384	384 robot episodes (scratch Gen-2 run)

IV. CALIBRATION

Wrist cameras. Kannala–Brandt (equidistant) fisheye models at full resolution, fitted with Kalibr on an 8×5 checkerboard of 25 mm squares (left f_x=789, right f_x=791 px). **Camera–IMU.** Kalibr [17] gives time offsets of 30.2 and 32.8 ms with mean reprojection errors of 6.0 and 5.6 px, which

is high and one reason pose accuracy is validated separately (Sec. V).

Camera to TCP. The TCP is the pinch centre with x along the approach direction, y along jaw closing and $z = x \times y$. The rotation is a fixed axis permutation ($x_{\text{TCP}} = z_{\text{cam}}$, $y_{\text{TCP}} = -x_{\text{cam}}$, $z_{\text{TCP}} = -y_{\text{cam}}$); the translation comes from caliper measurements (95 mm forward, 95 mm down) rotated by the camera pitch measured from IMU gravity (20.7°), giving $t = [0, 55.3, 122.5]$ mm, consistent to about 1 mm with a cube-based fit. Following UMI's use of a CAD constant, it was measured on one unit and applied to both.

Gripper width. Each frame takes the nearest gripper sample. The recorded normalized value was unreliable (the open end lay beyond the mechanical stop and the left direction flipped between sessions), so Gen-1 recomputes width from raw ticks with a per-episode closed anchor and the device span (1136.6 / 1248.2 ticks), width $= \text{clip}(f, 0, 1) \times 67.5$ mm (aperture 67.5 ± 5 mm, estimated from video of 50 mm-cube grasps). Gen-2 uses a later caliper calibration (gcal1): piecewise-linear knots at 0/20/40/60/80 mm, normalized as $g = \text{clip}(w/80\,\text{mm}, 0, 1)$, 0 = closed.

Clocks and synchronization. IMU device time is mapped to host time by a least-squares slope with the intercept re-anchored at the 5th percentile of the residuals, because USB latency only ever adds delay; the camera–IMU offset is then applied. Cameras free-run at slightly different rates (30.003–30.026 Hz), so frames are paired by nearest capture time. For bimanual data the left wrist is the master clock; a row is valid if both wrist poses are valid and the right wrist and head frames are matched within 20 ms.

Robot side (single-arm work). The robot's fixed head C922 (the same camera and position as during HRA_red) was calibrated on the same checkerboard (48 frames, RMS 1.05 px, $f = 728.7$ px), and its pose in the robot base was found by touching four board corners with the right jaw tip (Kabsch alignment of FK positions and board PnP; residuals 2–6 mm). The robot wrist camera is calibrated from 13 board views (0.16 px RMS, $f = 651.4$ px), and its hand-eye transform is solved with four methods (including Tsai–Lenz [18]) that agree to 1–2 mm and 0.3° in board-in-base pose. The camera sits 169.5 mm behind the TCP and 76 mm off its axis, and its optical axis points 33° below the TCP x axis.

TCP is not the jaw tip. The IK stack's TCP lies about 82 mm ahead of the physical jaw tip along the TCP x axis (found by projecting FK points into the wrist image). Contact is therefore modelled by the URDF fingertip box (70–80 mm behind the TCP, ± 19.6 mm), whose lowest corner counts as contact.

V. Wrist Pose and Metric Scale

Monocular pose. UMI localizes all demonstrations in one ORB-SLAM3 map per scene. On our data this failed: maps built before a scene change did not relocalize, and on the marker benchmark below its position tracks reversed direction often (median reversal rate 0.24 versus 0.07 for the method we adopt). We run MASt3R-SLAM [16] (official code, pinned commit, a build fix only) on the RGB wrist video undistorted to a pinhole view, single-threaded, every frame, fixed seed. After global optimization every frame is re-expressed on its final keyframe,

$$T_{WC}(f) = T_{WK}^{\text{final}}(k)\, T_{WK}^{\text{track}}(k)^{-1}\, T_{WC}^{\text{track}}(f), \tag{1}$$

a Sim(3) chain; frames tracked during relocalization are invalid unless they later became keyframes. Fig. 3 shows typical trajectories.

Scale 1: IMU visual–inertial. The SLAM trajectory is metric up to a per-episode scale s. With $R_{wb} = R_{wc} R_{cb}$, accelerometer samples f_b box-averaged around each frame, and the camera and lever-arm accelerations a_{cam}, a_{lever} obtained as Savitzky–Golay second derivatives (order 3, window W) of the SLAM camera position and of $R_{wc} t_{cb}$, each frame contributes three rows of

$$R_{wb}\, f_b - a_{\text{lever}} = s\, a_{\text{cam}} - g, \tag{2}$$

solved for s and $g \in \mathbb{R}^3$ by Huber-weighted iteratively reweighted least squares over the longest valid run. Both sides pass the same filter, so nothing is integrated twice. A scale is valid if $s > 0$ and $|g| \in [8.81, 10.81]$ m/s^2.

Pose benchmark. 14 demonstrations (28 hand tracks) with four 80 mm ArUco markers in view, run on the same GPU as the dataset; ground-truth scale is the median ratio of marker to SLAM displacement over frame pairs ten frames apart. $W = 15$ without bias was selected on a fit split of 9 demonstrations; on the 5 held-out demonstrations (10 tracks) MASt3R-SLAM with IMU scale gives median tracking coverage 1.00 versus 0.03 for ORB-SLAM3 with a stored map, scale error p50/p90 of 4.6/17.0% (versus 18.5/23.8% for one global constant), rotation error p50/p90 0.84/2.13° and a UMI-style 16-step relative position error p90 of 11.3 mm. The demonstrations themselves have no ground truth.

Scale 2: a cube of known size. For slow, smooth motions the accelerometer barely constrains the IMU fit: across the 200 HRA_red takes the IMU scale has a median of 0.012 and is zero or negative, which is physically impossible, in 95 of them (Fig. 4b). The red cube (edge 38 mm) is static. In frames where it is visible we segment it in HSV, fit its silhouette with two- or three-face models, or a single-face model within 35° of head-on, and solve PnP. Since the cube's centre $c_i = p_i + k\, R_i t_i$ must be constant, the inverse scale follows in closed form,

$$k = -\frac{\sum_i (p_i - \bar{p}) \cdot (u_i - \bar{u})}{\sum_i \|u_i - \bar{u}\|^2}, \qquad u_i = R_i t_i, \quad s = 1/k, \tag{3}$$

with MAD trimming and 200 bootstrap resamples. A take needs at least 10 PnP frames, a bootstrap spread of at most 10% and a cube-centre residual of at most 2 cm. On the HRL80 stacking takes, where the IMU scale is reliable, 37 of 57 right-hand episodes are PnP-valid and the PnP/IMU ratio has a median of 1.062 (p16–p84 0.959–1.274; Fig. 4a).

Scale 3: the table plane at a shared origin. In HRA_A100 the fingertip touches the plate during the origin hold, so the first MASt3R keyframe sees the table at a known distance.

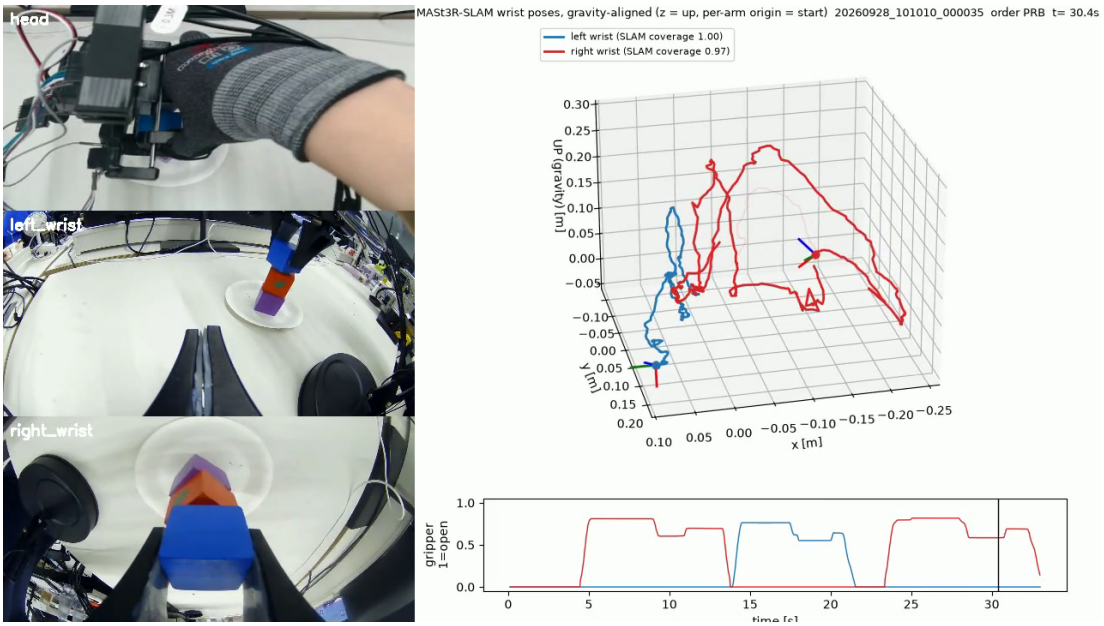


Fig. 3. MASt3R-SLAM wrist trajectories of an HRL80 stacking take (order PRB). Left: head camera and both wrist fisheye cameras. Right: trajectories gravity-aligned (z up), each arm's origin at its start pose, current tool axes drawn (x red, y green, z blue), metric scale from the IMU; below, both grippers (1 = open). Videos of three takes (PRB, RBP, PBR): `media/mast3r_trajectories/hrl80_0000{35,55,60}_wrist.mp4` in the repository.

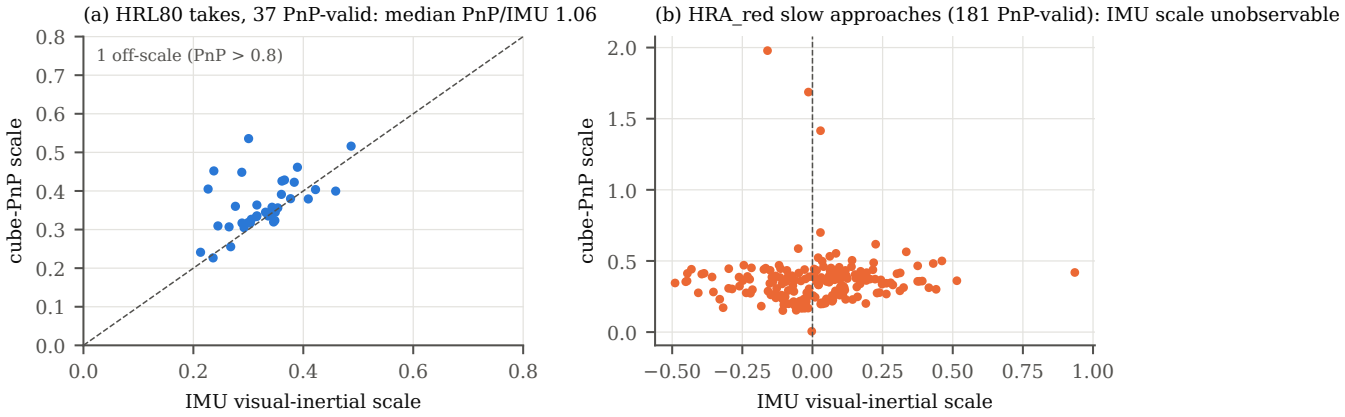


Fig. 4. Per-episode metric scale of the MASt3R-SLAM trajectory from the IMU and from cube PnP. (a) HRL80 stacking takes, where the IMU scale is reliable. (b) HRA_red approach takes: the IMU scale scatters around zero.

Excluding the finger, up to three planes are fitted by RANSAC and the largest one parallel to the plate is taken as the table; $s = |n \cdot t_{\text{tip}}|/(s_0\,|d|)$ with t_{tip} the fingertip in camera coordinates and s_0 the keyframe's Sim(3) scale. Against cube PnP on 40 takes the ratio has a median of 0.999 and a per-episode error of p50 8%, p90 37%. Each episode uses cube PnP where it is valid and the plane otherwise (35 and 72 of 107); the IMU is not used.

Silent pose jumps. Comparing ego and robot state statistics showed per-row state jumps above 100 mm in the ego data that the robot data does not have. Two causes were mixed. First, an exported dataset contains only training rows, so "adjacent" frames can be seconds apart: 1,482 of the 2,598 apparent jumps above 30 mm (57%) are such time gaps, and adjacency has to be decided from row timestamps. Second, the remaining jumps are real: MASt3R-SLAM occasionally moves the camera by 10 cm or more within one 30 Hz frame without flagging tracking loss (55 raw steps faster than 3 m/s in 24 training episodes, none preceded by a loss flag). The filter flags a step above 100 mm or 45° per 66.7 ms row, or an isolated step above 50 mm between two steps slower than 0.3 m/s. If the pose returns, the event is a spike and only its samples are invalidated; otherwise it is one-way and the rest of that arm's track is dropped, because a start-anchored state would be wrong from then on.

Camera–IMU consistency. Sessions whose gyro and SLAM angular velocities disagree (median cosine near 0 against 0.92–0.98 elsewhere) are quarantined. For the approach

TABLE II
GEN-1 PROCESSING FUNNEL FOR THE 349 STACKING EPISODES (COUNTS RECOMPUTED FROM THE CENSUS FILES IN THE REPOSITORY).

Stage	Episodes	Segments
Recorded (KEEP)	349	–
Pose + gripper exported (698 tracks)	349	–
Sync $\leq$20 ms, $\geq$65-frame window	284	1,306
− workspace / metric scale / IK		−677 / −50 / −18
$\geq$1 bimanual segment (training pool)	259	561
train / val (by episode, seed 0)	233 / 26	504 / 57

TABLE III
RETARGETING VARIANTS ON THE 1,229 CANDIDATE SEGMENTS OF THE 259 EPISODES, MEASURED AGAINST 120 HELD-OUT ROBOT EPISODES. NN: NEAREST-NEIGHBOUR JOINT-SPACE DISTANCE TO ROBOT POSTURES (RAD, P50/P90); W1: WASSERSTEIN DISTANCE OF $|\Delta q|$ PER STEP.

Variant	Usable	NN p50 / p90	Start clusters	W1 $\lvert\Delta q\rvert$
v1 single anchor	36.5%	1.49 / 1.55	1	0.00197
v2-K multi-seed IK	66.6%	1.21 / 1.60	–	0.00275
v2-Kr + posture penalty	67.0%	0.44 / 0.74	–	0.00310
v2-TR window retrieval	56.1%	0.27 / 0.45	100	0.00208

data, the exported camera–IMU rotation of the right wrist turned out to be wrong by about 18° about the camera x axis (cube-axis error 17.8→3.4° after correction); IMU tilt statistics computed before the correction are wrong by up to that amount, while training poses whose scale comes from cube PnP were unaffected.

VI. CONVERTING HUMAN MOTION INTO ROBOT DATA

A. Generation 1: joint-space retargeting

Segments and gates. Synchronized valid rows are cut into non-overlapping 65-frame segments (2.17 s). Per arm, a segment needs a valid metric scale, at least 95% of rows inside the robot workspace (a target is inside if its nearest neighbour among 6,000 robot teleoperation TCP poses is within 50 mm and within 20° in orientation), and an IK solution on every row. A segment is bimanual if both arms pass. Each TCP trajectory is mapped to a reBot B601 6-DoF arm by least-squares IK on its URDF. Absolute workspace geometry is not recoverable across episodes, so each segment is re-anchored at one fixed robot start pose (the median active pose of our robot teleoperation data); motion is transferred, absolute cube positions are not. The jaw encoder becomes a binary gripper command.

Funnel. Table II lists what each gate removes. 65 episodes fail the $\leq$20 ms synchronization requirement for a full 65-frame window; the remaining 284 yield 1,306 candidate bimanual segments, of which 677 leave the robot workspace, 50 fail the metric-scale check and 18 fail IK, leaving 561 (43.0%). Two sessions (6 episodes) were quarantined for an IMU–camera rotation inconsistency; none contributed a segment. 47% of segments start in the first 5 s of an episode.

Split. The 259 episodes with at least one bimanual segment are split at random by episode (seed 0, 10%): 233 train / 26 val, i.e. 504/57 segments and 32,760/3,705 frames. No episode spans both sides. The split is not stratified; validation has 3–6 episodes per order, 22 main / 4 early, from 9 of the 13 non-quarantined sessions.

Retargeting v2. The single start anchor was the main source of loss: every segment began at one robot posture, so most left the workspace and the retargeted trajectories covered one posture cluster. v2 replaces it with multi-seed retargeting under a strict budget: every real-robot quantity (seed bank, workspace gate, posture penalty, trajectory windows) comes from 30 robot episodes only, and results are measured against a disjoint set of 120 robot episodes. A seed bank of 2^{17} Sobol samples is filtered by workspace, table floor, manipulability and inter-arm clearance ($\geq$75 mm), leaving 1,195 seeds. Variant K rolls out the best 8 seeds under a cost on position, rotation, joint velocity and acceleration; Kr adds a posture distance to robot data; trajectory-window retrieval (TR) starts each human segment from the closest feasible robot trajectory window (64 nearest by a 96-D relative-motion feature). Table III compares them on the same 1,229 candidate segments. TR is the most robot-like: posture distance to the held-out robot data falls from 1.49 to 0.27 rad (median) and start postures span 100 clusters instead of 1. It solves fewer segments than the kinematic variants because it cannot place fast motions; among the 686 segments both TR and Kr solve, their joint-velocity distributions are similar (W1 0.00207 vs 0.00226), so the posture gain, not a dynamics gain, is what TR adds. The remaining failures are almost all "no workspace-compatible window" (399 of 402 unsolved).

Robot-like re-collection. To test whether the loss is a collection-style problem we recorded HRL80 (60 episodes, 10 per order) with the robot-like protocol; its interpretation rules were fixed before retargeting. Live rotation speed halved (median over-threshold fraction 0.144 to 0.067), but the per-episode TR yield did not change (0.59 vs 0.59) and TR usable rose only from 56.1% to 58.9%; the per-episode correlation between live robot-likeness and TR yield is weak (Spearman $\rho = -0.11$, $p = 0.05$, pooled $n = 316$). Workspace placement, not collection style, remains the main failure. The final TR dataset combines both collections: 996 segments (896 train / 100 val) from 316 source episodes, with the original 689 segments preserved bit-identically.

B. Generation 2: the Cartesian contract (CART20)

Retargeting to robot joints had two costs: the workspace and IK gates removed a third to a half of the candidate motion, and the joint-space target differed from the Cartesian action space our robot policies are fine-tuned in. Gen-2 converts each human wrist trajectory directly into the relative end-effector representation of the robot teleoperation data, so that ego and robot samples share one tensor contract.

Contract. Rows are stored at 15 Hz. For row time t and each arm the action is a chunk of 16 future poses

$$A_k = T(t)^{-1}\, T(t + k\Delta), \qquad k = 1, \ldots, 16, \tag{4}$$

with $\Delta = 50.05$ ms (the robot data's step, 0.80 s in total), current-anchored rather than sequential, and interpolated on the raw 30 Hz track (linear in position, SLERP in rotation, only between valid samples no more than 50 ms apart). Each arm contributes position, a 6-D rotation and a gripper value (20 dimensions for two arms, "CART20"); the model's 32-D action is padded with 12 dimensions that are exactly zero and never supervised. The state is each arm's pose relative to its own task-start pose, $T(t_0)^{-1}T(t)$, plus both gripper values, the same layout as the robot state (there is no shared left–right SLAM frame). The gripper is the jaw aperture divided by 80 mm; the largest observed value is 0.847, so "fully open" is never reached. A row is kept only if both arms are valid at t, all 16 targets are valid and a head-camera frame exists within 20 ms; there is no padding and no workspace or IK selection. Eleven synthetic tests pin the contract.

Data. The source is the 273 original stacking episodes with valid per-hand IMU scale and 60 HRL80 episodes, of which 3 are refused because one hand's IMU scale is invalid, leaving 330. Of 111,533 rows, 57,283 (51.4%) are usable; the rest fall on tracking losses, synchronization gaps or the last 0.8 s of an episode. The split is frozen from Gen-1 (validation = the 26 held-out episodes plus HRL80 episodes 49–54): 298 train episodes with 51,612 rows and 32 validation episodes with 5,671 rows. The pose-jump filter (Sec. V) acts on 44 events in 42 training episodes and removes one episode entirely; training shrinks to 297 episodes and 48,411 rows (-6.20%), validation is unchanged. The largest step between time-adjacent rows falls from 422.5 to 90.5 mm and steps above 100 mm from 18 to 0; 7 raw steps faster than 3 m/s remain inside valid rows.

How ego data differs from robot data. On 60 ego and 60 robot episodes, with ego poses attached to a robot start pose: the robot's IK reaches 99.7% (left) and 99.5% (right) of ego positions, and 72.3%/74.6% of full ego poses. The IK configuration used on the robot, which locks one wrist joint, reproduces only 45.2%/52.2% of the robot's *own* recorded poses, a deployment limitation that exists without any human data. Ego hands use a smaller workspace (state p95 0.221/0.214 m versus 0.395/0.369 m) with similar 0.4 s translations (p95 93/99 versus 91/100 mm), and both ego hands are still at the same time far more often (19.5% versus 2.4% of samples). The largest difference is visual: the variance of the image Laplacian, a sharpness measure, has a median of 1,854/1,487 for the ego wrist fisheye views and 59/64 for the robot's wrist cameras.

Robot-like wrist rendering. To narrow the visual gap, a fisheye frame can be re-rendered as a virtual pinhole camera with the same orientation, so labels stay valid: horizontal field of view 62–70°, principal point shifted down so that the jaws sit low as on the robot, then downscaled, blurred, JPEG-compressed and photometrically jittered. ROBOT100 applies this to every frame.

C. Single-arm variant and a shared task frame

Right-only contract. The tensor layout is unchanged: the left state is the identity pose with gripper 0, the left action is zero motion, and a 20-D loss mask supervises only the right arm's position and rotation (9 dimensions; the jaw is unused). The task string is "Approach to the red cube".

Robot-camera rendering (robotcam v2). The first version (v1) rendered the wrist view with the guessed camera above. The measured robot C922 has a 52° horizontal field of view with a centred principal point, so v1 showed the cube much later in the approach than the robot camera does. In v2 every robot-camera pixel's ray is projected into the fisheye model and resampled, and labels are expressed as robot-TCP poses through the hand-eye transform (robot TCP $\rightarrow$ UMI TCP: $-33°$ about y, $t = (-94.8, 2.4, 38.4)$ mm).

Task frame G. A cross of tape on the head-camera base plate, about 35 cm from the robot gripper, is the task origin. The robot's right jaw tip touched three points on the tape line; they are collinear to within 0.38 mm over a 60.4 mm baseline and agree in height within 2 mm. G has its origin at the cross centre, x along the tape towards the human side and z along base z: origin $(59.0, -74.9, 6.2)$ mm in the robot base, heading 47.09°. With the 35 mm plate the table is at $z = -27.1$ mm. On the human side the origin pose is the median position and mean rotation over the origin hold; its yaw is measured per take by matching table features around the cross against a reference take (error p50 0.21°, p90 1.17°), with tilt from IMU gravity. Assuming a mean yaw instead had produced false rejections and a 57 mm endpoint-height spread (35 mm with measured yaw). The hold itself repeats to 14 mm (p50; p90 37 mm) and 6.2° (p50), with the camera pitched 63.7° down.

Table-aware conversion. Expressing human motion in G and starting it from a robot start pose can push the fingertip into the table. The trajectory is therefore warped from its own start towards the robot start pose with smoothstep weights that decay to zero over the take, on separate schedules for horizontal position, height and rotation, so the human endpoint near the cube is kept. A height correction c is then solved per take as a quadratic program,

$$\begin{aligned} \min_{c} \; & \textstyle\sum c^2 + 50(\Delta c)^2 + 2000(\Delta^2 c)^2 + 60\, m\,(c-g)^2 \\ \text{s.t. } & z_{\text{tip}} + c \geq z_{\text{table}} + 5\,\text{mm}, \end{aligned} \tag{5}$$

with $c_0 = 0$, an endpoint lift of at most 30 mm and g pulling low frames towards 12 mm clearance; the result is replayed through IK. Takes are classed PASS (lift $\leq$20 mm), PASS-CORRECTED (20–30 mm) or rejected (IK success below 98%, missing origin hold or yaw, hold moved more than 10 mm). Table IV shows the variants. CT4 decays position, height and rotation on separate schedules; CT5 adds measured origin yaw, the corrected IMU gravity, absolute state in G and the hold gate; CT6 moves the robot start 6 cm back and 4 cm down. CT5 and CT6 keep 86 takes with fingertip clearance p50 12.6 mm and minimum 7.6 mm; most rejections are the first session (before the origin hold existed) or takes whose origin yaw could not be measured.

TABLE IV
TABLE-AWARE CONVERSIONS OF HRA_A100. ACCEPTED = PASS + PASS-CORRECTED. CLEARANCE: FINGERTIP ABOVE THE TABLE AFTER CORRECTION, ACCEPTED TAKES.

Variant	Takes	Accepted	Main rejections	Clearance p50 / min
C (mean yaw)	107	81	lift >30 mm 10, old 7	19.5 / 7.0 mm
CT4 (per-axis decay)	107	87	lift 10, old 7	20.6 / 7.9 mm
CT5 (yaw, G, hold gate)	107	86	yaw not measured 9, old 7	12.6 / 7.6 mm
CT6 (start back/down)	107	86	as CT5	12.6 / 7.6 mm
CT7 (low start, +27)	134	90	IK 19, yaw 11, old 7	12.4 / 7.6 mm

VII. MODEL AND TRAINING

Model. X-VLA [13] through LeRobot [14], from the public `xvla-base` checkpoint: a Florence-2 [19] vision-language encoder, a 24-layer soft-prompted transformer (hidden 1024, 16 heads) and a flow-matching [15] action head; 879M parameters, all trainable. Inputs are the three views (head, left wrist, right wrist, resized to 224 px), the instruction and the proprioceptive state. Training draws t, forms $x_t = \varepsilon t + a(1-t)$ and regresses the clean action; inference integrates from noise in 10 steps by default. For Gen-2 the public head (20-D action and state) was widened to a 32-D action and 20-D state, with existing rows copied exactly and the widened model checked to reproduce the base.

Which soft prompt? X-VLA conditions on a dataset/embodiment id that selects one of 30 sets of 32 soft prompts and domain-specific state/action projections. Our first runs used id 0. Reading the public base weights shows that only slots 10–17 were ever trained: in the other 22 slots the encoder and decoder biases are exactly zero and the prompt weights stay near their initialization. Ids 0 and 6, both used by us, were therefore untrained slots. The trained slots are not a better start either: the initial loss on R312c is 1.11 for slot 0, 1.12 for 6, 1.61 for 15, 2.35 for 10, 2.65 for 16, 4.07 for 17 and 22.80 for 11. All Gen-2 runs use the unused slot 20 as our embodiment's own slot.

Gen-1 recipe. For state time t the model predicts a 30-step chunk $a_i = q_{t+5+i} - q_t$, $i = 0, \ldots, 29$, of cumulative joint displacements of the retargeted pseudo-joints (a 5-frame lead, so the chunk covers 0.2–1.17 s ahead), plus the absolute gripper. The state is 14-D (6 joint angles and a binary gripper per arm). Two auxiliary terms tie joint space to the measured human motion: (C) the relative TCP position of the measured trajectory in spare action dimensions (weight 2.0) and (D) a forward-kinematics loss between $\mathrm{FK}(q_t + \hat{a}_i)$ and the measured TCP position (weight 20). Targets are measured human transitions, never relabelled robot commands. Optimization: 300k steps, batch 4, AdamW (lr 10^{-4}, the vision-language group at $0.1\times$, weight decay 10^{-4}, 1k warm-up, cosine decay to 2.5×10^{-6}), gradient clipping 10, fp32, one RTX 4090, a checkpoint every 10k. The checkpoint selection rule (minimum geo score, Sec. VIII) was fixed before training.

Gen-2 recipe. MSE on action dimensions 0–19 only; dimensions 20–31 are zeroed at the transformer's input and output at every denoising step, in training and at inference, and a preflight checks that padding targets are ignored and padding gradients vanish. Batch 8, 8-bit AdamW (policy learning rate 10^{-4}, warm-up 1k, cosine decay ending exactly at the last step), bf16 mixed precision, seed 1000. For the single-arm data, masked targets are replaced by the detached prediction before subtraction and the loss is normalized by the number of supervised elements; a 300-step smoke test verified that masked and padded dimensions receive exactly zero gradient. Fine-tuning starts a fresh optimizer and uses the robot dataset's normalization statistics; co-training draws exactly four ego and four robot samples per batch.

Operations. Jobs run on a Ray cluster pinned to an RTX 4090 or RTX 5090 node (their PyTorch versions differ, so cross-node runs are not bit-identical). Launchers refuse to start on a busy GPU, a code-hash or domain-id mismatch, a failed gripper-contract gate or low disk space; a short smoke run with save and resume precedes each main run; and an archiver copies every checkpoint to external storage with SHA-256 verification before deleting it from the node.

VIII. EVALUATION PROTOCOL

Gen-1 offline. On the 26 held-out episodes we take 7 start frames from each of the 57 validation segments (every 5 frames in 0–30), giving 399 samples and 798 arm-samples, and compare the predicted chunk with the recorded future. For horizons $k \in \{1, 4, 8, 16, 30\}$ we report the TCP position error between $\mathrm{FK}(q_t + \hat{a}_k)$ and the measured TCP (median, mm), the joint MAE and that of a *zero-motion* predictor ($\hat{a} \equiv 0$), and the cosine between predicted and measured TCP displacement. The *geo score* is the mean of the median TCP error over the five horizons. Because most samples are near-static, every metric is also reported on the *moving* subset, arm-samples whose measured TCP moves $\geq$20 mm by k=30 (187 of 798, 23.4%). This gate was added after the 60k checkpoint had been evaluated and is used for reporting only.

Prompt swap. To test instruction use, the final checkpoint is queried on every sample with each of the six order instructions under one noise seed, and with the true instruction under two further seeds. The spread of the six predictions measures the instruction's effect; the spread across seeds is the sampling-noise floor. **Paired bootstrap**: episode-level resampling (10k draws, identical for both models) of the relative difference in geo score between two initializations.

Gen-2 diagnostics. Teacher-forced direction cosine on robot frames (predicted versus recorded translation at k=4/8/16 for moving frames), repeated under all left/right swaps and axis sign flips; an image/state domain swap; optical-flow versus pose-axis correlation of the wrist videos as a convention check; wrist sharpness as the variance of the Laplacian; training-loss comparisons as window means and ratios; and, for the single-arm data, held-out loss (the masked training loss on all validation rows).

What the splits do not test. Validation episodes share operator, scene, sessions and all six orders with training, and layouts are unlabelled. Results therefore measure within-distribution generalization to new *episodes*, not to new layouts, orders or scenes. Every number in this report is recomputed

from evidence files in the repository by a claim-verification script.

IX. Results: Joint-Space Stacking (Gen-1)

Table V and Fig. 5 summarize the evaluation of the model fine-tuned on the 561 Gen-1 segments.

(1) Aggregate errors mostly reflect static phases. The geo score reaches 2.78 mm at 160k and 2.88 mm at 300k and is flat after about 120k (Fig. 5a). On moving samples the error is an order of magnitude larger: 11 mm at the shortest horizon and 40 mm at k=30, with a p90 of 90 mm; the moving-subset geo score is 23.0 mm versus 2.9 mm on all samples.

(2) On moving samples the policy predicts the direction of motion and beats a no-motion baseline. The direction cosine rises from 0.72 at k=1 to 0.90 at k=30, and joint MAE is 13–32% lower than the zero-motion predictor across horizons (6.59 versus 8.68° at k=30; Fig. 5b).

(3) On all samples, the policy is worse than predicting no motion. Joint MAE is about twice the zero-motion MAE at every horizon (0.30 versus 0.15° at k=1): during static phases the policy predicts small spurious motion, and the median predicted-to-measured displacement ratio is 1.41 at k=1. On a robot this would appear as creep before grasps or after placement; residual pose noise in near-static windows is a plausible, untested contributor. At k=30 the spread of predictions is 0.77 of the spread of targets, so the long horizon is under-dispersed rather than collapsed, and fast or unusual motions are pulled towards the mean.

(4) The order instruction changes predictions but does not improve them. In the prompt swap (Fig. 5c), the six instructions move the predicted TCP by a median of 4.8 mm at k=30 on moving samples, about 50× the sampling-noise spread (0.09 mm) but only 8% of the predicted displacement (62 mm). The true instruction is not closer to the recorded motion than the wrong ones: its error is 39.9 mm versus 39.5 mm for the mean wrong instruction, and its mean rank among the six is 3.55 (chance 3.5; rank 1 in 16% of cases, chance 17%), with the same picture at every horizon and on all samples. The prediction with the true instruction and the original seed reproduces the reference evaluation to within 10^{-5} mm. Much of a 2.2 s segment is order-independent (reaching, lifting), and once stacking has begun the order is partly visible in the scene, so nothing in training forces the model to use the instruction; a test restricted to choice points needs more samples than 26 validation episodes provide. Per-order breakdowns (3–6 validation episodes per order) are too small to rank orders.

Data lost before learning. 57% of the candidate segments are removed; the workspace gate accounts for 677 of the 745 removals (91%), and in 531 of those (78%) only one arm leaves the robot workspace. Because the gate removes motions by their extent, the kept set is likely biased towards compact motions; we have not quantified this bias. Retargeting v2 (Table III) recovers much of this loss.

Ego pretraining for robot fine-tuning. The ego checkpoint above and the base checkpoint were each fine-tuned on R150 with the same recipe, data and evaluation; only the initialization differs. At the matched step of 250k, the ego-initialized model has a 10.0% lower geo score (episode-level paired bootstrap, 10k draws: 95% CI -14.7 to -5.6%), a 7.2% lower moving-sample geo score (CI -13.4 to -1.8%) and 6.8% lower moving joint MAE at k=30 (CI -12.7 to $+0.5\%$); 8 of the 10 evaluation episodes favour it. The evaluation episodes are part of both fine-tuning sets, so this measures how well each model fits the robot training distribution, not generalization. A comparison on R90 was stopped early with only two matched steps (10k, 20k); there the signs are mixed (all-sample geo -33% and -11%, moving-sample geo $+2\%$ and $+13\%$ for the ego initialization), so we draw no conclusion from it.

Pretraining on the final TR dataset. X-VLA was also pretrained on the 996-segment TR dataset with the same recipe (planned 300k steps). The run was stopped at 211.7k steps; the 100k and 200k checkpoints are kept, and no robot fine-tuning result from them exists.

X. Results: Cartesian Transfer (Gen-2)

All runs in this section use the Gen-2 recipe; robot data is R312c.

Ego-only model on robot frames. After 40k steps on ego data alone, the model was queried, teacher-forced, on 120 robot frames and compared with a model trained 40k steps on robot data (these frames are in that model's training set, so its value is an upper reference). The median cosine between predicted and recorded translation at 0.4 s is $+0.81/+0.77$ (left/right) for the robot model and $-0.05/-0.03$ for the ego model. No combination of a left/right swap and axis sign flips raises the ego model above $+0.16$, and the wrist-camera optical flow has the same sign relations to TCP motion in both domains, so this is not a convention error.

The gap is in the images. Table VI swaps the two inputs. With ego images the model keeps a cosine of 0.69–0.90 whatever state it is given, robot state included; with robot images it falls to between -0.25 and $+0.49$ whatever the state. The failure is visual, which fits the 30-fold sharpness difference of the wrist views (Sec. VI).

Ego pretraining as an initialization (312 episodes). We fine-tuned on R312c from the base model (A) and from the ego model after 50k steps (B), with identical data, schedule (150k-step cosine), seed and recipe; both were stopped by us near 60k. At step 200 the training loss is 0.945 for A and 0.244 for B. The ratio B/A is 0.49 over the first 5k steps, 0.875 over 5k–20k and 0.95 over 20k–40k, and over 55k–60k the two are tied (0.0733 versus 0.0745; B lower at 13 of 24 logged points) (Fig. 6). Ego pretraining therefore speeds up early fitting of the robot data and gives no measurable advantage in training loss by 60k. This is training loss with one seed per arm; we have no held-out or closed-loop comparison of A and B. Two 300k runs finished later, but on different robot datasets (R312c for the ego-initialized run, R384 for the scratch run), so they are not a transfer comparison.

Attempts to close the visual gap. Applied to every frame (ROBOT100), the robot-like rendering of Sec. VI drops the

TABLE V

GEN-1 OFFLINE HELD-OUT RESULTS (26 EPISODES). TCP ERROR: MEDIAN FK POSITION ERROR (MM); MAE: JOINT MAE (DEG), WITH THE ZERO-MOTION PREDICTOR IN PARENTHESES; COS: TCP DIRECTION COSINE. 160K IS THE PRE-REGISTERED SELECTION, 300K THE FINAL STEP.

		horizon k (steps of 1/30 s after a 5-frame lead)				
Subset	Metric	1	4	8	16	30
All (798)	TCP err., 160k	1.3	1.7	2.2	3.1	5.7
	TCP err., 300k	1.5	1.8	2.2	3.2	5.7
	MAE, 300k	0.30 (0.15)	0.34 (0.19)	0.42 (0.25)	0.57 (0.36)	0.97 (0.51)
Moving (187)	TCP err., 160k	11.2	15.0	19.9	29.2	39.1
	TCP err., 300k	11.2	15.0	19.4	29.3	39.9
	MAE, 300k	2.02 (2.32)	2.68 (3.36)	3.29 (4.38)	4.30 (6.34)	6.59 (8.68)
	cos, 300k	0.72	0.80	0.84	0.86	0.90

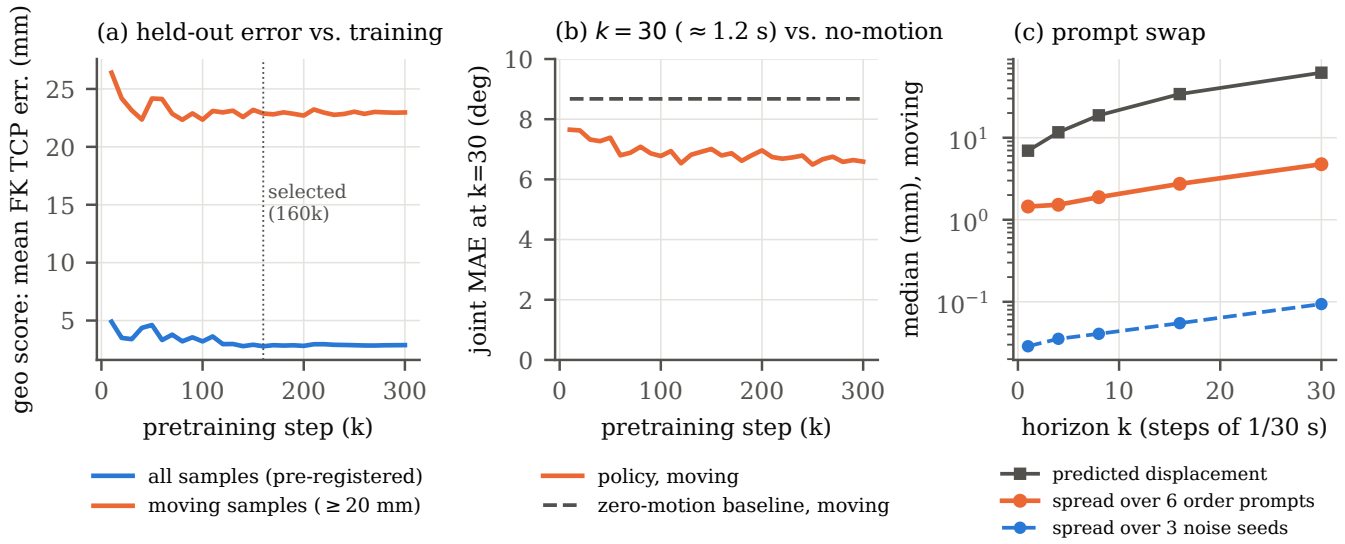


Fig. 5. Gen-1 offline evaluation. (a) Geo score over training on all versus moving held-out samples; the pre-registered rule selects 160k. (b) Joint MAE at k=30 on moving samples, policy versus zero-motion predictor. (c) Prompt swap on the final checkpoint: prediction spread across the six order instructions (same seed) versus across three noise seeds (true instruction).

TABLE VI

EGO-ONLY CHECKPOINT (40K), MEDIAN DIRECTION COSINE AT k=4/8/16 (LEFT | RIGHT) WHEN IMAGES AND STATE COME FROM DIFFERENT DOMAINS. 100 EGO VALIDATION FRAMES AND 100 ROBOT FRAMES; 21–55 MOVING FRAMES PER CELL.

Images + state	Left k=4/8/16	Right k=4/8/16
ego + ego	0.87 / 0.91 / 0.92	0.89 / 0.94 / 0.84
ego + zero	0.77 / 0.80 / 0.89	0.85 / 0.80 / 0.66
ego + robot	0.76 / 0.81 / 0.90	0.72 / 0.69 / 0.72
robot + robot	0.36 / 0.30 / −0.01	0.19 / 0.02 / −0.15
robot + zero	0.17 / 0.25 / −0.25	0.24 / 0.06 / −0.02
robot + ego	0.16 / 0.24 / −0.16	0.49 / 0.33 / −0.03

wrist sharpness median from 1,854/1,487 to 181/132, still two to three times the robot's 59/64. A run with 70% of frames re-rendered was stopped at 5k steps; a co-training run with four ego (ROBOT100) and four robot samples per batch was stopped at 88.6k steps without evaluation. A 300k ROBOT100 pretraining run (297 episodes, 48,411 rows, 49.6 epochs) finished on 5 October at a training loss of 0.007; its checkpoints are the ego initializations of the next experiment. No offline transfer metric was computed for the re-rendered models.

Low-data ablation: 30 robot episodes. The 312-episode comparison gives the robot data every chance to wash out the initialization; the question that matters for using human data is whether it helps when robot data is scarce. We fine-tuned on R30 with R312c normalization statistics, seed 1000 and a 600k-step cosine schedule from three initializations: the base model (scratch) and the ROBOT100 run at 100k and at 300k steps. The 100k checkpoint comes from the middle of the same 300k schedule, so its learning rate had not decayed; it is not a separate 100k-step run. The arms ran on different machines (scratch and 100k-init on an RTX 4090 with PyTorch 2.6, 300k-init on an RTX 5090 with PyTorch 2.7.1), so they are not bit-identical twins.

Table VII and Fig. 7 show the 312-episode pattern, stretched out. Both ego initializations start at less than half the scratch loss (window ratio 0.54 and 0.49 over 1k–5k). The 100k-step

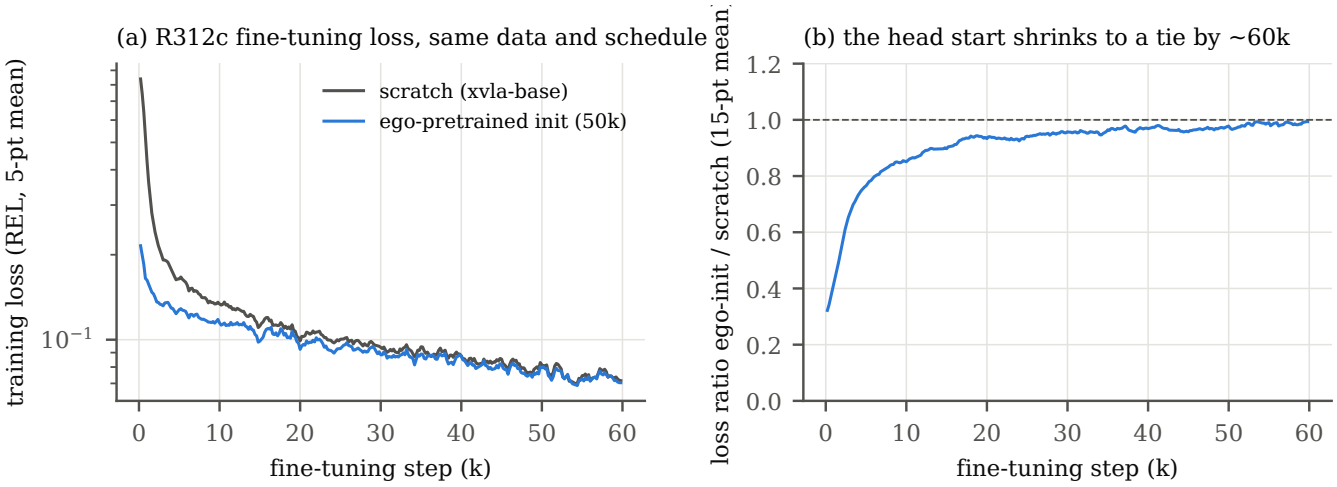


Fig. 6. R312c fine-tuning from the base model versus from the 50k ego checkpoint, same data, schedule and seed. (a) Training loss. (b) Ratio of the two; values below 1 favour the ego initialization. Training loss only.

TABLE VII
R30 FINE-TUNING, TRAINING LOSS (200-STEP RUNNING MEAN) AT MATCHED STEPS, AND THE RATIO OF WINDOW MEANS TO SCRATCH (BELOW 1 FAVOURS THE EGO INITIALIZATION). ONE SEED PER ARM; NO HELD-OUT OR ROBOT EVALUATION.

Step	Scratch	Ego-PT 100k init	Ego-PT 300k init
200	0.801	0.239	0.446
1k	0.312	0.103	0.117
10k	0.065	0.056	0.040
50k	0.019	0.017	0.014
100k	0.009	0.009	0.007
Window		ratio	ratio
1k–5k		0.544	0.493
10k–20k		0.823	0.697
50k–100k		0.970	0.856
100k–150k		1.021	0.914

initialization loses its advantage by about 100k steps (ratio 0.97 over 50k–100k, 1.02 over 100k–150k); the 300k-step one keeps a smaller advantage longer (0.86 and 0.91) and reaches parity at about 200k. Beyond about 150k all losses are 0.001–0.004 at the printed precision and ratios are not meaningful. The 100k-step initialization starts lower than the 300k-step one (0.239 versus 0.446 at step 200) although it was pretrained less; we have not explained this. All three runs were stopped by us to free GPUs, at 471k, 305k and 310k, after 214, 139 and 141 epochs: with 30 episodes every arm fits its training data almost exactly long before the schedule ends. Training loss here measures how fast each initialization memorizes the robot data, not how well it acts; the relevant comparison is closed-loop at early and middle checkpoints (roughly 10k–150k), and no R30 checkpoint has been run on the robot yet.

XI. RESULTS: SINGLE-ARM APPROACH

Stacking showed that, once both domains share one action contract, the remaining gap is visual and geometric. A single-arm approach to a cube isolates that gap with far fewer

TABLE VIII
SINGLE-ARM HELD-OUT LOSS (MASKED MSE ON THE 9 SUPERVISED DIMENSIONS) BY CHECKPOINT; TRAINING-SUBSET LOSS IN PARENTHESES.

Dataset (val eps)	5k	15k	30k	45k/50k	105k
HRA_red v1 (15)	–	0.150 (0.066)	0.198 (0.053)	–	0.299 (0.018)
A100 A: start (11)	0.124 (0.110)	0.125 (0.058)	0.141 (0.031)	0.224 (0.020)	–
A100 B: origin (11)	0.132 (0.097)	0.147 (0.057)	0.174 (0.031)	0.190 (0.022)	–

confounds: one arm, one object, no grasp, no task order, and success can be judged by eye.

HRA_red (v1). Of 200 takes, 19 fail the cube-scale gates (14 with too few PnP frames, 5 with too large a spread) and 15 fail a physical sanity check on the scaled trajectory (scale between 0.1 and 1.0, SLAM travel at least 0.8 of the PnP-observed approach distance, travel at most 0.8 m). The remaining 166 are split into 151 train episodes (20,319 rows) and 15 validation episodes (2,011 rows); the median scale is 0.354 (p5 0.188, p95 0.500) and the median cube-centre residual 0.17 cm. With the guessed-camera rendering, the 300k-step run finished on 5 October at a training loss of 0.002 (118 epochs). Held-out loss rises from the first evaluated checkpoint (Table VIII, Fig. 9a): 0.150 at 15k (training subset 0.066), 0.198 at 30k, 0.213 at 60k, 0.265 at 90k and 0.299 at 105k, while the training-subset loss falls to 0.018. With 151 episodes the model overfits almost immediately.

First robot runs and what they exposed. v1 checkpoints from 15k to 75k ran 8,397 control cycles (658 run starts) on 4 October, and 100k, 200k and 300k ran 61 planned sessions on 6 October while the chunk-execution settings were tuned; the operator's verdicts were "accuracy dropped a lot" and "too much zigzag". From a start near the cube the arm moved towards it; for cubes far to the left it dragged along the table, by about 39 and 26 cm in two runs. Interpreting these runs required knowing where the human data sits relative to the robot, which v1 never did: its labels are relative motions of

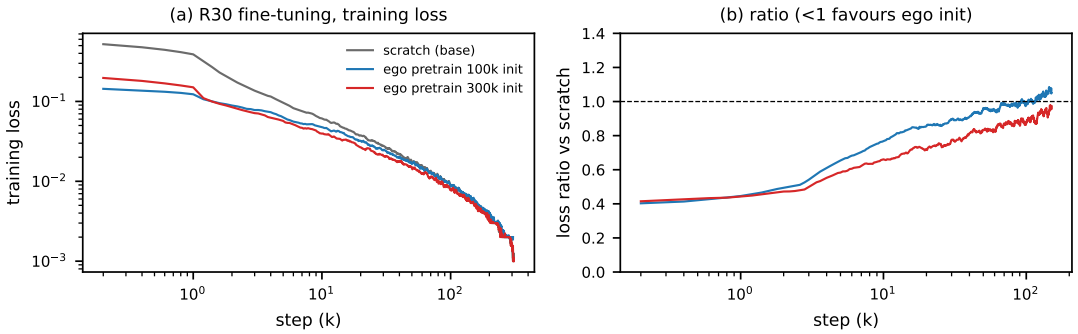


Fig. 7. R30 fine-tuning (30 robot episodes) from the base model and from ego pretraining at 100k and 300k steps. (a) Training loss (log–log, smoothed). (b) Ratio to scratch, shown to 150k; beyond that the 3-decimal losses make the ratio noise. Training loss only, one seed per arm.

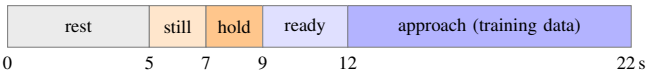


Fig. 8. HRA_A100 take cycle: rest, jaw tip still on the taped cross, recording starts with a 2 s origin hold, ready, 10 s approach. The origin hold places the HandUMI jaw tip on the same taped cross that the robot jaw tip touched during calibration.

the UMI tool frame and its start state is the human's own start. Linking takes to the robot base after the fact (cube in the head camera, wrist trajectory through cube PnP, tilt from IMU gravity) linked only 64 of 166 takes, with start heights spread over 4 cm and up-axis error p50 10°, p90 28°, and it exposed the 18° camera–IMU error of Sec. V. The human wrist camera at the take start and the robot wrist camera at a reachable start cannot coincide: matching the human's view would need the shoulder joint past its limit; the closest feasible start (pose A) puts the camera within 1–14 mm horizontally but 78 mm higher and 8° steeper. Re-reading v1 200k from pose A with the jaw-tip model under three tool-frame conventions gave: tool axis unrotated, the tip ended about 2 cm from the cube and 3 cm above the table; rotated by the 33.1° camera pitch, it overshot by about 11 cm; rotated and offset by the camera–TCP translation, it ended about 4 cm from the cube and 10 cm above the table. These are single runs reconstructed from forward kinematics, not measurements of success. A run on robotcam v2 reached about 64k steps (training loss 0.026) before being stopped.

HRA_A100: a shared origin. Each take is an automatic cycle (Fig. 8): 5 s rest; the operator places the HandUMI jaw tip on the taped cross and holds it still for 2 s; recording starts with a further 2 s hold (the origin); a 3 s "ready" phase moves the hand to a robot-like start pose; at "go" a 10 s approach is recorded, and training data starts at "go". Cube placements were planned as 40 centre, 20 left/right, 20 front/back and 20 edge. On 6 October 107 takes were kept in four sessions (7, 8, 9 and 83 takes; the first session predates the origin hold). Scale comes from cube PnP or the origin plane, frame and yaw from G (Secs. V–VI).

Start-anchored versus origin-anchored state. On 93 takes (82 train, 11 validation; the excluded 14 are the old session, 7 takes in which the hand moved over 20 mm during the hold, and one physics failure) we trained two otherwise identical models: A anchors the state at the hand pose at "go", B at a fixed origin frame F (the cross, with the tape heading and the median hold pitch). Both use robot-TCP labels and the robotcam v2 rendering. Their held-out losses are close and both are lowest at the first checkpoint (Table VIII, Fig. 9b): A 0.124 and B 0.132 at 5k, rising to 0.224 (A, 50k) and 0.190 (B, 45k). With 82 training takes, both overfit from 5k steps; neither anchoring is better on this measure.

Start pose. We compared ten candidate robot start poses derived from the median and medoid of the human "go" poses in G, moved back, right and down; the operator preferred the lowest (pose 08, TCP $(273, -168, 116)$ mm, joint margin 9.1°). On 7 October 27 further takes were recorded with a lower human start (origin to "go" about 89 mm instead of about 247 mm); from pose 08 only 5 of them pass, so they were dropped, and the current run (CT7-old, 85 takes from 6 October, start 08, 100k-step schedule) uses only the 6 October data. CT5 and CT6 runs were stopped after 7k and 3k steps while settings changed; there is no validation split for CT6 and CT7.

XII. Deployment and Robot Observations

Inference and control. The policy runs on a Mac (Apple-silicon GPU, bf16) with 5 (bimanual) or 10 (single arm) denoising steps. The robot cameras are captured at 640×480, resized to 224×224 and normalized as in training; in single-arm mode the left wrist image is black. The state follows the training contract (TCP poses from FK of the measured joints, relative to an anchor taken at run start or to a fixed anchor file), and absolute targets $T_k = T_{\text{now}} A_k$ are solved by Pink [20] IK on the URDF. In *PLAN mode* one forward pass is kept in flight and IK-solved rows (50.05 ms apart) stream to the robot service, which plays them as 100 Hz knots in MIT impedance mode; a planned knot jumping more than 90° makes the arm hold. Optional real-time chunking [21] pulls each new chunk towards the previous chunk's not-yet-executed targets.

Settings that mattered. (i) *Chunk execution*: successive 16-step chunks are blended. With an exponential blend of

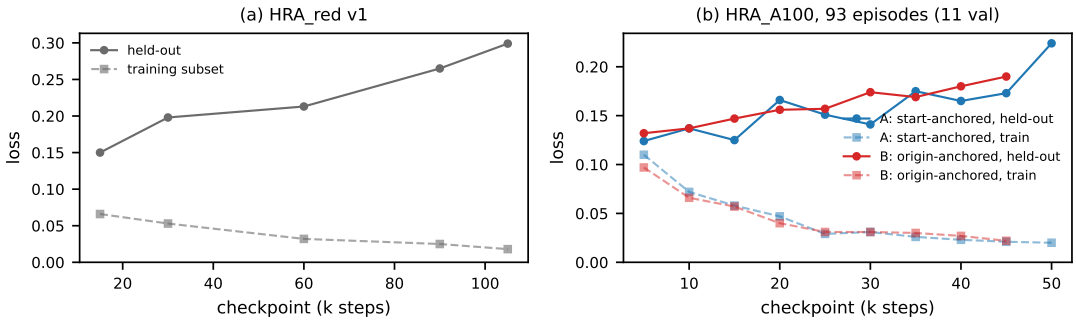


Fig. 9. Held-out (solid) and training-subset (dashed) loss by checkpoint. (a) HRA_red v1, 15 validation takes. (b) HRA_A100, start-anchored (A) and origin-anchored (B) state, 11 validation takes. All runs overfit from the earliest checkpoint.

0.25, no cross-fade, no smoothing and a minimum of one executed row per chunk, motion was acceptable; with the code defaults (0.35, 4 rows, 3°, 6 rows), which a restart of the interface silently restored on 7 October, the operator reported that "inference quality had dropped a lot" until the values were restored. Behaviour on the robot depends on settings that are not part of the checkpoint; the loader now restores them. (ii) *Tool-frame convention*: models trained on robot-TCP labels (robotcam v2, A100) must run with zero camera pitch and zero camera–TCP offset; the 33.1° pitch and offset apply only to v1, and anchored models must be served with the same anchor file (F for B, G for CT5–CT7). (iii) *Cube-size stop*: after the arm was seen pushing the cube instead of stopping, the interface measures the red cube's apparent size ($\sqrt{\text{area}}$ in pixels on the raw 640×480 wrist frame) every 0.1 s and stops when two readings exceed a threshold (170–250 px were used); in follow mode it resumes when the cube shrinks below 85% of the threshold. It is a proximity proxy, not verified contact. The wrist image was also zoomed 1.5× because the cube looked smaller than in training at the same position. (iv) *Guards*: during v1 the interface stopped a run after 450 mm net travel, 80 mm between inferences, a still period, or an output above 250 mm or 60°; a 6 s limit ended runs after 1–4 cycles and was removed. For A100 models all numeric guards are off and the left arm and jaws stay locked.

Stacking on the robot. Gen-2 checkpoints were executed on the reBot through the deployment interface: 71 checkpoints, 49,431 control cycles and 1,681 run starts between 2 and 4 October. The cycle log records IK tracking of the model's own waypoints, not task outcome, so it yields no success rate. Two operator observations were not logged. The ego-initialized model at 10k fine-tuning steps grasped cubes but picked the wrong colour. The ego-initialized model fine-tuned on R312c for 300k (B300, checkpoints between 75k and 205k) completed the full three-cube stack in closed loop. The number of trials was not recorded and no video was taken, so we state only that closed-loop three-cube stacking has been demonstrated, with no success rate. It also says nothing about the value of ego pretraining: no scratch model trained on R312c was evaluated under the same protocol. Of the B300 checkpoints that stacked, only 150k and 200k survive a storage clean-up, so the demonstration can be re-run only from those. A trial logger (checkpoint, robot data, instruction, layout, stack level 0–3, wrong colour, failure reason) was added on 4 October but has not been used.

Approach on the robot. The planned-execution log records 192 sessions on 6 and 7 October. On 6 October, 65 sessions (v1 checkpoints, the chunk-setting tuning above) were all stopped by the operator except one, which the interface closed when a planned row jumped by 90.9°. On 7 October, 127 sessions ran: B 45k (3), A 50k (6) and CT5 5k (4), then B 50k with the 1.5× zoom (114 sessions, many of them follow-mode resumes); 69 of the 127 ended on the cube-size stop and 58 were stopped by the operator. The log has no outcome field, so we cannot say how many approaches reached the cube. The cube appears orange in the fisheye-rendered training images and red on the robot camera, a colour gap we have not corrected.

XIII. Corrections and Known Inconsistencies

Several of our own intermediate readings were wrong, each found by an independent check: the robot start tilt was 25°, not 3°; the robot's wrist view is about 1.32× magnified and 11° lower than the human's, not 2.5×; "labels drive the tip 5 cm into the table" was the TCP–tip offset; the table height moved from −54.3 to −5.9 to −27.1 mm as the contact model was fixed; IMU tilt statistics were off by up to 18°; a mean origin yaw caused false rejections; a camera-orientation convention error in an earlier export corrupted rotations until a frame-regression test found it; and two of our embodiment ids pointed at untrained soft prompts.

Writing this report also exposed inconsistencies in our code, which we list rather than hide. (1) An inter-arm clearance check is used only in retargeting v2, not as a Gen-1 gate. (2) In the Gen-1 target code a binarized gripper label 1 means *open*, although a comment says closed. (3) Three gripper calibrations are in use (Gen-1 ticks with a 67.5 mm aperture, the gcal1 caliper knots for Gen-2, and the collector's open/closed files); each dataset names its own. (4) The camera–IMU time offsets used by the exporter differ in sign and magnitude from the raw calibration output, and the conversion step is not recorded. (5) Two different right fisheye calibration versions are used by the exporter and by one viewer tool. (6) The IK step thresholds are

documented per 15 Hz row but applied on 30 fps rows. (7) In PLAN mode the bimanual path does not pass gripper widths to the waypoint solver, so the jaws would likely stay closed; the single-arm runs, which hold the jaws, are unaffected. (8) Several diagnostics were run from scripts outside the repository and their outputs are stored as captured logs, not re-run. (9) Fitting scripts for the G frame, the fingertip box and the 18° camera–IMU correction are not in the repository; only their outputs are.

XIV. Lessons

Each of the following was found by an explicit check rather than by a loss curve. (i) *Every pose or scale source has a regime in which it fails silently*: ORB-SLAM3 across scene changes, MASt3R-SLAM on rare one-frame jumps, the IMU on slow motion, cube PnP on oblique single-face views and the origin plane on a few takes (p90 error 37%); only a second, independent reference shows the failure. (ii) *A pretrained model's conditioning interface has to be audited, not assumed*: two of our embodiment ids pointed at soft prompts that had never been trained. (iii) *When human and robot data share a tensor contract, the remaining gap can be localized by swapping inputs between domains*; here it is the wrist images, not kinematics or normalization. (iv) *A shared physical origin is cheaper than post-hoc registration*: linking takes after the fact reached 64 of 166 with centimetre errors, while a taped cross touched by both robot and human gives a common frame with a sub-millimetre line fit. (v) *The TCP of the IK stack is a convention, not the fingertip*; contact reasoning needs fingertip geometry. (vi) *Training-loss comparisons of initializations end in a tie whenever the robot data is fitted to near zero*, at 312 and at 30 episodes alike, and small single-task sets overfit at once; only held-out or closed-loop evaluation at early checkpoints can show a benefit. (vii) *Execution settings are part of the policy*: blend and smoothing parameters that live in the interface changed robot behaviour more than a change of checkpoint, and outcome logging has to exist before the robot runs, not after.

XV. Limitations and Future Work

Stopped early, no conclusion: the R90 fine-tuning comparison (two matched steps), the 70%-re-rendered pretraining (5k steps), the ego/robot co-training run (88.6k steps, never evaluated), the robotcam v2 run (64k) and the CT5/CT6 runs. **Not yet done:** robot fine-tuning from the 996-segment checkpoints; any robot or held-out evaluation of the R30 arms; a held-out or closed-loop comparison of ego-initialized and scratch models; outcome logging on the robot.

Not supported by this study: (i) a closed-loop success rate for stacking or approach, or any closed-loop difference between ego-initialized and scratch models; (ii) generalization to new cube layouts, since layouts were not recorded and Gen-1 discards absolute geometry by design; (iii) generalization to unseen stacking orders, since all six orders appear in training; (iv) transfer beyond one operator, one scene and a few days; (v) any benefit of human data for robot learning beyond fitting the robot training distribution faster; (vi) that the ego-only policy or the re-rendered wrist images transfer to robot cameras; (vii) colour grounding in the approach data.

Planned experiments: log every robot trial (model, checkpoint, robot data size, start pose, layout, instruction, stack level 0–3 or reached yes/no, failure and stop reason) with fixed execution settings, and run 20–30 trials per model; compare scratch and ego-initialized models at matched fine-tuning steps and at R30 for the step and data budget at which stacking first succeeds; test early approach checkpoints (5k–15k) of A, B and CT7 from the same starts and cube positions; record layouts from the head camera and hold out layout regions; hold out two orders entirely and test instruction following; correct the colour of rendered wrist images; train with a static-phase-aware loss; and use the approach prior as the first segment of the stacking policy.

More broadly, this work is a first step towards egocentric human demonstrations as scalable supervision for robot policies: learning human-to-robot representations that keep task geometry and intent while discarding embodiment detail, and evaluating them with predictive world models before execution. The present results show that the bottleneck is as much in pose recovery, filtering, calibration and evaluation design as in the policy.

XVI. Conclusion

We built an upstream open-source hand-worn gripper with an added wrist IMU, collected 349 bimanual stacking demonstrations and two single-arm approach datasets, and turned them into robot training data in two ways. Retargeted to robot joints, the stacking data teaches a 0.9B VLA the direction of held-out human motion (cosine 0.90 at $k{=}30$ on moving samples) but not the use of the order instruction, and static-phase drift is hidden by aggregate metrics; multi-seed trajectory retrieval removes much of the retargeting loss. Converted to the robot's own Cartesian contract, whole episodes survive without feasibility filters once silent SLAM jumps are removed, and the remaining human-to-robot gap is visual and lies in the wrist images. As an initialization, ego pretraining speeds up early robot fine-tuning at 312 and at 30 robot episodes, and the training-loss advantage disappears as the robot data is fitted. An ego-initialized model has completed full three-cube stacks on the robot, so the task is solved in closed loop; whether ego pretraining reaches that point with fewer robot episodes or steps is the open question. For slow single-arm motions, a known object and a shared physical origin replace IMU scale and bring human data into the robot's frame with sub-centimetre consistency; models trained on 80–150 such takes overfit quickly and move towards the cube from nearby starts. Measured success rates, spatial and task-order generalization, and a benefit on disjoint robot data require the controlled experiments listed above.

APPENDIX A
CODE MAP AND REPORT HISTORY

Code organisation. The repository is one installable Python project. Each pipeline stage is a package under `src/` with its own README and characterization tests (`src/<pkg>/tests/`): capture `src/handumi_collector/`, pose `src/mast3r_pose/`, conversion `src/ego_cart20/` (Gen-2) and `src/rebot_prep/` (Gen-1), training `src/training/`, deployment `src/rebot_deploy/`; machine-specific data roots are defined once in `src/ego_paths/`. `scripts/` holds only the launchers for the GPU nodes and the robot computer, grouped by stage; `configs/` the calibrations; `results/` the records behind every number; `archive/` keeps superseded and exploratory code verbatim, with an index (`archive/README.md`). On 5 November 2026 the main-path code was restyled without behaviour change: tests recorded on the old code pass unchanged (`docs/refactor_2026-11.md`).

Code map (paths after the 5 November 2026 refactor; old-to-new map in `docs/refactor_2026-11.md`): collector and protocols `src/handumi_collector/` (`collector/autoloop.py`, `collector/recorder.py`, `collector/integrity.py`); hardware and IMU firmware `hardware/`, `firmware/`; upstream licence, NOTICE and our patch `upstream/`; export and sync `src/mast3r_pose/export_episode.py`, `scripts/pose/`, `src/rebot_prep/census/`; MASt3R-SLAM wrapper and scales `src/mast3r_pose/` (`run_perframe.py`, `imu_metric_scale.py`), `src/ego_cart20/cube_pnp.py`; Gen-1 `src/rebot_prep/pseudo_joint/`, `src/rebot_prep/retarget/`, and its dataset builders, trainer and offline evaluator `archive/gen1_c8old/`; Gen-2 `src/ego_cart20/` (contract, pose filter, converter, validators); single arm `archive/hra_single_arm/`; training `src/training/`, launchers `scripts/train/`; evaluation and diagnostics `analysis/`, `archive/rel16_audit/`; deployment `src/rebot_deploy/`, `scripts/deploy/`; claim verification `analysis/verify_claims.py`.

Repository and follow-up. The laundry and garment work that followed this project, recorded with the same hand-worn grippers, continues in its own repository, `egocentric-garment-manipulation`, and is not part of this report. The capture packages (`handumi_collector`, `ego_collector`, `umi`) started as copies shared with that repository; here they are trimmed to the modules this pipeline imports.

Report history. Version 2 (30 September 2026) covered Gen-1 stacking; version 3 (4 October) added Gen-2 and transfer diagnostics; version 4 (7 October) added the R30 ablation and split the work into Part I (stacking), Part II (approach) and a methods document; version 4.1 (12 October) added the hardware description. Version 5 (12 October) merged them into one report without changing any result; the earlier documents remain in the repository. Version 5.1 (1 November) updated repository paths after the code restructure. This version 5.2 (5 November) updates the code map to the refactored layout and adds the code-organisation paragraph; no result changed.

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